

Reference Architecture RA-02 — Predictive Maintenance at 1,000 Sensor Channels, Hybrid with AWS-Hosted Services

Audience: VP of IT/OT and the architecture review board · **Status:** For approval **Companion document:** RA-01 — 100% on-premises. Section numbering is identical for side-by-side comparison; sections that are unchanged from RA-01 say so and state only the delta. **Version basis:** SUSE Edge 3.6.1 release matrix, verified against documentation.suse.com on 2026-07-23 · **Re-verify by:** 2026-10-23

1. Executive summary

Decision requested: approve the same two-phase deployment as RA-01 (pilot → site rollout, §13) **plus** a bounded set of AWS-hosted services for cross-site visibility, long-horizon analytics, and off-site disaster recovery — under a closed-list egress contract in which **raw process telemetry, recipes, and trade-secret data never leave the site**, and the OT control plane never depends on the cloud.

What changes vs. RA-01, in one paragraph. The plant floor is byte-for-byte the same architecture: the same HA edge platform scores every channel on-site, and Class-C1 data (raw traces, recipes) still has **no egress path**. What this variant adds is a second landing zone — an AWS account under our control — that receives only an enumerated schema of **derived** fields (§6.2): health scores, RUL forecasts, attribution ranks, pseudonymized tool IDs. That buys three things RA-01 cannot offer: (1) fleet-wide dashboards when this pattern extends to additional sites, (2) long-horizon model analytics and retraining on derived features without touching site compute, and (3) ransomware-resistant off-site escrow of client-side-encrypted backups (keys stay on-site — AWS stores ciphertext it cannot read). The trade is a defined, auditable derived-data flow and a WAN dependency for those three functions only; **the site runs fully autonomously when the WAN is down** (§8).

Situation, complication, and business case are those of RA-01 §1 (sourced: McKinsey 30–50% downtime reduction; Deloitte +25% productivity / –70% breakdowns / –25% maintenance cost; Siemens *True Cost of Downtime 2024*; ITIC 2024) — with one addition: as the pattern scales to multiple sites, per-site on-prem duplication of dashboards/analytics grows linearly, while this variant centralizes exactly the layer that is safe to centralize. Choose RA-01 where policy demands zero WAN egress of any kind; choose RA-02 where fleet visibility and off-site DR justify a derived-data-only conduit. The documents are otherwise the same platform — approving RA-02 includes everything in RA-01.

2. Scope, assumptions, and claim labeling

Identical to RA-01 §2 (labels, A1–A6), plus:

#	Assumption	Sensitivity
A7	Derived-data egress volume ≤ 1 msg/channel/min summary cadence $\Rightarrow \leq \sim 17$ msg/s site-wide, trivially within a single VPN tunnel's capacity.	Low
A8	The data-classification ruling in §6.2 (which fields may cross) is signed off by site security and legal before Phase 1 exit — it is a business decision this document records, not an architect's assumption.	Critical
A9	AWS region [FILL], single account per environment, no cross-region replication of derived data unless DR policy requires it.	Medium

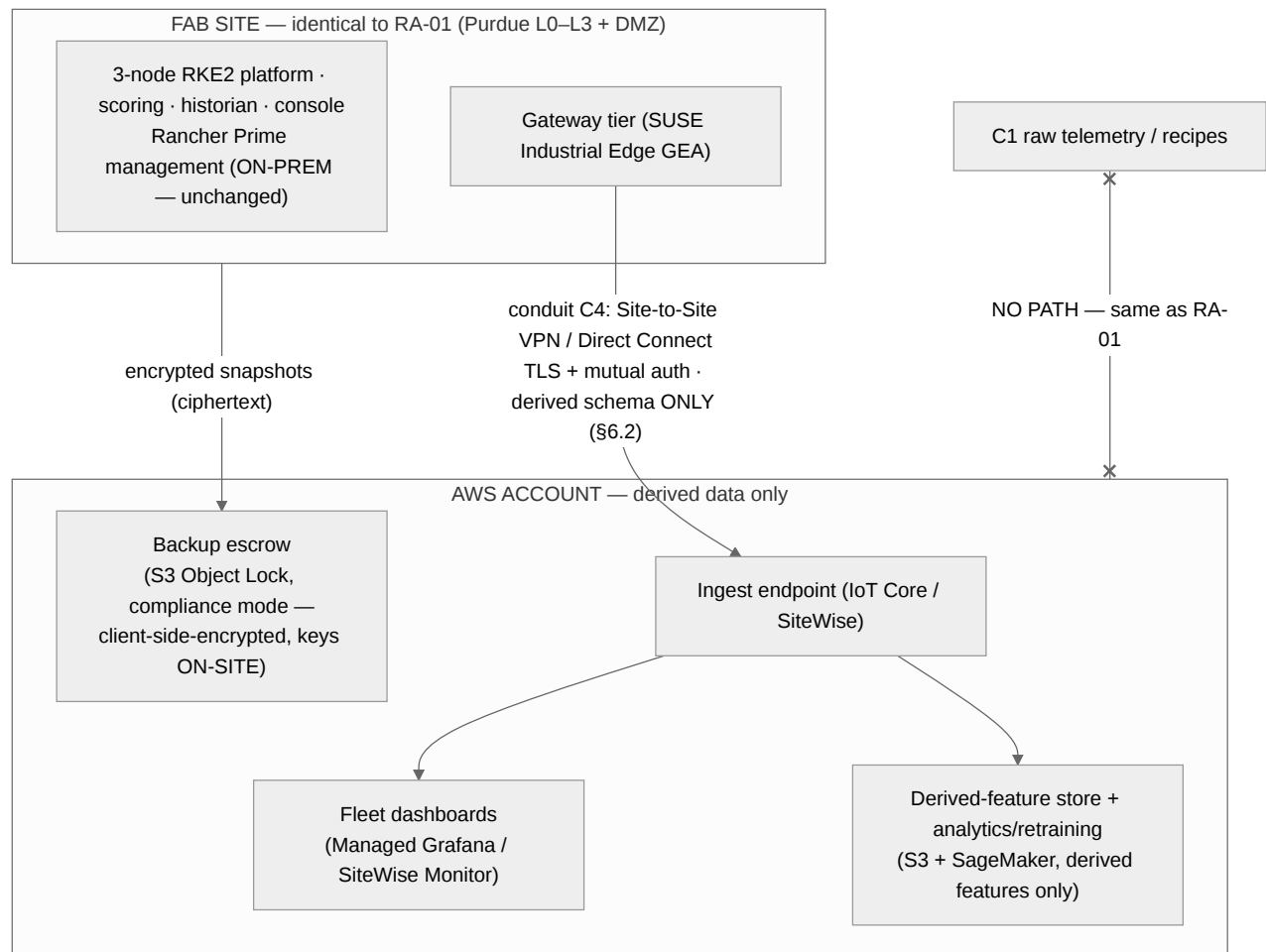
3. Requirements

RA-01 §3 applies unchanged, with these NFR additions/overrides:

NFR	Target	Verification
Data sovereignty	Zero egress of C1 data (unchanged). Derived C2 egress only per the closed schema §6.2, via one governed conduit	Quarterly egress audit now includes VPC flow-log review on the AWS side
WAN-loss autonomy	Site scoring, alarming, console, and storage fully operational with WAN down; cloud dashboards go stale and recover by backfill from gateway/broker buffers	Pilot disconnect drill (§13)
Cloud tenancy security	AWS account hardened per §7.6 (SCPs, KMS with on-site key escrow for backups, no public endpoints, IAM least privilege)	Third-party cloud config review [FILL: schedule]

4. Architecture overview

4.1 Zone model — RA-01 §4.1 plus one governed conduit



The one structural rule of this document: Rancher Prime management stays **on-prem** (as in RA-01). AWS never hosts the OT control plane; it hosts derived-data services and ciphertext escrow — a closed list (§4.5). The OT loop has zero cloud dependency.

4.2–4.4 Tiers, design properties, brownfield

Identical to RA-01 §4.2–4.4. The gateway tier's governed-egress design, dormant in RA-01, is **active** here: the GEA is the single component permitted to publish to conduit C4, and only the §6.2 schema [Lab-verified mechanism: the kit's gateway forwards derived-only fields and withholds raw structurally; single-node scope].

4.5 The AWS closed list (anything not listed is out)

Service	Function	Data class received
IoT Core / SiteWise ingest	Terminate conduit C4; validate schema	C2 derived only

Service	Function	Data class received
Managed Grafana or SiteWise Monitor	Cross-site fleet dashboards	C2
S3 (derived store) + SageMaker	Long-horizon analytics; model retraining on derived features; retrained parameters return to site via the DMZ mirror path	C2
S3 Object Lock (compliance mode)	Immutable off-site escrow of client-side-encrypted etcd/Longhorn/config backups — WORM that even the root account cannot shorten [Sourced: AWS docs]; decryption keys never leave the site	C3 ciphertext
KMS, CloudTrail, SCPs, IAM	Account guardrails and audit	—

Prior art for this shape exists and is cited rather than reinvented: *AWS Guidance for Industrial Data Fabric, Predictive Maintenance with Amazon Monitron* [Sourced: AWS Solutions Library]; SiteWise Edge's own documented pattern is local processing with only approved aggregates sent to the cloud — with the honest caveat that anything computed at the edge and sent up **is** in the cloud, which is exactly why §6.2 is a closed, signed schema [A8].

5. Component catalog, alternatives, and exit paths

RA-01 §5 applies unchanged on-site. Additional cloud-side rows:

Function	Selected	Alternative	Exit path
Derived ingest	AWS IoT Core/SiteWise	Self-hosted MQTT bridge on EC2; or RA-01 (no cloud)	MQTT standard; repoint the gateway's egress URL
Fleet dashboards	Managed Grafana	Grafana on-prem (RA-01 §5 console covers single-site)	Dashboards are Grafana JSON — portable
Analytics/retraining	SageMaker on derived features	On-prem GPU node (RA-01 Tier 3 hardware)	Training code is ours; features are open Parquet
Backup escrow	S3 Object Lock	Second on-site failure domain (RA-01 §10.4); any S3-compatible WORM store	Ciphertext blobs — provider-agnostic
Site ↔ cloud conduit	Site-to-Site VPN (Direct Connect at fleet scale)	—	Standard IPsec/BGP

Deliberately not selected — AWS Outposts: it keeps stored data on-prem, but its service link carries control-plane traffic, updates, and telemetry to the parent region and the management plane resides in-region [Sourced: AWS Well-Architected data-residency lens]. That re-introduces a cloud dependency inside the site boundary that this design exists to avoid; recorded honestly in §14.

6. Data governance & sovereignty

6.1 Classification — identical to RA-01 §6.1.

6.2 Egress matrix and the wire schema (the centerpiece artifact)

RA-01's matrix applies with one added row:

Flow	Content	Direction	Allowed?
Gateway → AWS ingest (conduit C4)	Only the schema below	L3 → AWS	Yes — TLS, mutually authenticated, schema- validated at both ends

The complete wire schema (closed; field-level opt-out; anything else is a violation to be blocked and alerted):

```
{
  "site_id": "pseudonymized site token",
  "tool_id": "pseudonymized tool token (mapping table stays on-site)",
  "window_end": "timestamp (see note)",
  "health": "0-100 score",
  "state": "HEALTHY|WATCH|WARNING|CRITICAL",
  "rul_cycles": "integer or null",
  "attribution": ["ranked sensor-role tokens, not raw channel names"],
  "alarm_count_window": "integer"
}
```

Note stated plainly for the review board: even pseudonymized tool IDs plus timestamps can leak production-schedule intelligence in aggregate. The pseudonymization mapping lives only on-site; cadence is coarse (per-minute summaries [A7]); and the schema itself is subject to the signed classification ruling [A8] with field-level opt-out. This mirrors the governed-egress contract the reference kit implements and demonstrates [Lab-verified mechanism: derived-only forwarding with raw structurally withheld; single-node scope].

6.3 Backup escrow crypto contract

Backups are encrypted **on-site** before transfer (site-held keys; envelope encryption); S3 Object Lock compliance-mode retention makes them immutable against ransomware and insider deletion [Sourced]. AWS can return the bytes; it can never read them. Key custody procedure [FILL: site key-management standard].

7. Security architecture

RA-01 §7.1–7.5 apply unchanged on-site (defense-in-depth stack, IdP/RBAC, supply chain, patching, 62443 SL-T 2 mapping). Additions:

7.6 Cloud tenancy hardening

Dedicated account per environment; SCP guardrails (deny public S3, deny regions outside [A9], deny KMS key export); no public endpoints — ingest reachable only via the VPN/DX conduit; IAM least-

privilege with roles, no long-lived keys; CloudTrail + VPC flow logs shipped to the site SIEM; quarterly cloud-config review [FILL].

7.7 Conduit C4 controls

Mutual TLS with site-issued client certs on the gateway; schema validation on both ends (gateway refuses to serialize off-schema fields; ingest rejects and alerts on any); NeuVector/firewall rules restrict the egress source to the gateway service identity alone — everything else on-site still has **no** route to the WAN.

8. Availability & failure modes

RA-01 §8 applies unchanged, plus:

Failure	Behavior	Recovery	Target
WAN loss (VPN/DX down)	Site fully autonomous: scoring, alarms, console, storage unaffected. Cloud dashboards stale; gateways buffer derived summaries (65,000-msg buffer each [Sourced]) and backfill on reconnect	Automatic backfill	No site impact; dashboard staleness = outage duration
AWS region/service event	Same as WAN loss from the site's perspective; escrowed backups unaffected (Object Lock)	Await region recovery	No site impact
Conduit compromise suspected	Kill the VPN at the site firewall — one rule; site continues as RA-01	Forensics on flow logs; re-key	Site unaffected by design

This table is the argument for the structural rule in §4.1: because the OT loop has no cloud dependency, every cloud failure degrades **visibility**, never **production**.

9. Sizing & bill of materials

Site BOM identical to RA-01 §9. Cloud-side: modest managed-service consumption driven by A7 (~17 msg/s derived ingest, GB-scale monthly derived store, backup escrow at snapshot cadence) — cloud cost is operational, not capital, and is quoted at procurement [FILL: AWS pricing at chosen region; the derived-only volumes keep this in the low hundreds of \$/month range, stated as an assumption-driven estimate, not a quote].

10. Day-2 operations

RA-01 §10 applies unchanged, plus:

- **10.7 Cloud ops calendar additions:** monthly — SCP/IAM drift review, VPN cert expiry check; quarterly — restore-from-escrow drill (prove the ciphertext + on-site keys actually restore),

egress-schema audit against flow logs, cloud cost review.

- **10.8 RACI additions:** cloud account ownership (R: platform eng., A: IT/OT VP, C: site SecOps + cloud CoE [FILL]); escrow key custody (R: site SecOps, A: CISO org).

11. Risk register

RA-01 R1–R7 apply. Additions:

#	Risk	L×I	Mitigation	Residual	Owner
R8	Derived-data schema creep ("just one more field") erodes sovereignty over time	M×H	Closed schema under signed ruling [A8]; change requires the same sign-off; quarterly audit vs. flow logs	Low	Site SecOps
R9	Aggregate derived data leaks schedule intelligence	M×M	Pseudonymization on-site, coarse cadence, field-level opt-out; noted openly in §6.2 rather than discovered later	Medium	Site SecOps
R10	Cloud account misconfiguration exposes C2 data	M×M	§7.6 guardrails; no public endpoints; third-party review	Low	Platform eng.
R11	Organization reads "hybrid" as license to send more later	M×H	This document IS the boundary: the closed list §4.5 + schema §6.2 are the approved envelope; anything beyond returns to the board	Low	IT/OT VP

12. Cost model

RA-01 §12 formula and inputs apply. Delta: add cloud operational cost (assumption-driven estimate §9, quote at procurement) to the annual-cost line; add the avoided cost of per-site dashboard/analytics duplication to the benefit line **only when** site #2 is approved [A6] — single-site economics should justify RA-02 on DR escrow + analytics alone, or RA-01 is the better choice and this document says so.

13. Implementation roadmap

RA-01 §13 phases and exit criteria apply, with these additions:

Phase	Added content	Added exit criteria
1 — Pilot	Stand up AWS account + conduit C4 + ingest; escrow pipeline	WAN-disconnect drill: site autonomous, backfill verified, zero loss; schema-violation test: off-schema field rejected AND alerted at both ends; restore-from-escrow drill passes with on-site keys; signed data-classification ruling in hand [A8]

Phase	Added content	Added exit criteria
2 — Rollout	Fleet dashboards live; analytics pipeline on derived store	Egress audit clean across one full ops-calendar cycle

14. Alternatives considered & exit strategy

RA-01 §14 applies, plus:

Option	Why not selected
RA-01 (zero egress)	The honest default. Choose it if fleet visibility and off-site escrow don't justify a derived-data conduit. RA-02 exists for the multi-site trajectory and DR posture — not because cloud is inherently better
AWS Outposts	Data residency yes, but control-plane/service-link dependency into the region contradicts the OT-autonomy rule (§4.1); stated per AWS's own data-residency lens, not vendor FUD
Cloud-hosted Rancher management (EKS)	Architecturally possible (agents dial out) but makes the OT fleet's management WAN-dependent — rejected; management stays on-site in both documents
Azure equivalent of this hybrid	Feasible with IoT Operations/ADX; not selected to keep one cloud landing zone and because the site-side platform is cloud-agnostic either way — the conduit and schema, not the cloud brand, are the architecture

Exit from AWS entirely: kill conduit C4, repoint escrow at any S3-compatible WORM store, run dashboards on-prem — i.e., become RA-01. That this sentence is short is the point of the design.

15. Appendices

15.1 Evidence base — identical to RA-01 §15.1, one addition

Artifact	Proves	Does prove
Kit gateway governed egress + offline buffer/flush (air-gapped counters, buffer/flush exercised)	The derived-only egress and disconnect-buffering mechanisms this document's conduit C4 and WAN-loss rows rely on	Fleet-scale disconnected operation; cloud-side behavior

15.2 Glossary — RA-01 §15.2 plus: Object Lock (S3 WORM immutability) ·

SCP (AWS service control policy) · DX (Direct Connect) · conduit C4 (the single governed site → cloud flow defined here).

15.3 References — RA-01 §15.3 plus: AWS IoT SiteWise Edge documentation

(local processing, aggregate-only egress pattern) · AWS S3 Object Lock documentation · AWS Well-Architected *Data Residency / Hybrid Cloud Services Lens* (Outposts service-link nuance) · AWS Solutions Library: *Guidance for Industrial Data Fabric on AWS*; *Guidance for Predictive Maintenance with Amazon Monitron*.

